

CPSC 2010: Introduction to Computer Science

Mutation April 17, 2026

Logistics

HW8 released, due May 3, 11:59pm

All late homework due May 4, 9am

Friday review sessions in DL 220, 3:30pm

Final exam May 6, 2pm, Davies Auditorium

Practice exam will be released by Monday, April 27

the **case** special form

To evaluate

```
(case <expr>
  [(<expr>) <expr>...]
  [(<expr>) <expr>...]
  ...)
```

1 Evaluate the scrutinee expression, then check if it is equal to the quoted test expression of each branch, starting at the top.

2 Evaluate the remaining expressions in the matching branch and return the value of the last one. If no branch activates, return nothing.

```
(define (describe-degree n)
```

```
  (case n
    [(0) "constant"]
    [(1) "linear"]
    [(2) "quadratic"]
    [(3) "cubic"])))
```

```
(describe-degree 2)
```

the **set!** special form

To evaluate

(set! <id> <expr>)

1

Find the binding for
<id> in the relevant
environment.

2

Evaluate <expr> and
change the binding for
<id> to the resulting
value.

```
(define n 6)
(define x 5)
```

TOP-LEVEL ENVIRONMENT (ACTIVE)

Identifier	Value
...	
*	built-in procedure #<procedure:*>
n	6
x	5
...	
If identifier not found, raise "unbound identifier" error.	

To evaluate

(set! <id> <expr>)

1

Find the binding for
<id> in the relevant
environment.

2

Evaluate <expr> and
change the binding for
<id> to the resulting
value.

```
(define n 6)
(define x 5)
(set! x (- x 1))
```

TOP-LEVEL ENVIRONMENT (ACTIVE)

Identifier	Value
...	
*	built-in procedure #<procedure:*>
n	6
x	4
...	
If identifier not found, raise "unbound identifier" error.	